GRK Test 1 · one-day sprint

Wed 11 Nov → light Thu 12 → test Fri 13 Nov 08:30 · 15 pts · lectures 1–6

- Each block: 5 min read the recipe (Tasks tab) → do the official example with the answer covered (pack Part B) → 2–3 random problems in Solve with the solution hidden → 10 min cards for that topic.
- Never look at a solution before you try. Mixing topics later (mock, cards on "All") is what makes it stick.
- Every task on paper: formula → table of steps → drawing → boxed answer, and write your convention.
- Tools: phone trainer (Today tab has this checklist with practice buttons) · printed pack Part B (model answers) · squared paper.

Wednesday 11 Nov · the whole day

✓	time	block
☐	08:30–09:00	**Cold pre-test** Mock tab → 30 min, before reading anything. Don't look anything up. It shows what you don't know yet, and trying first makes the rest of the day stick. Write down the tasks you couldn't do.
☐	09:00–09:50	**Numbers: displays + image** Quick, sure points. Learn the 4K slide answer (4096×2160 ≈ 10^7 ticks → $1.8 \cdot 10^9$ B/s), ns not µs, MB = 10^6, eye 2063/2069, bilinear P1, P2, N. *Types:* Data rate of a display · Time to display one pixel · Memory of a vector image · Memory of a bitmap · Linear resolution (ppi) of a display · Maximum useful resolution (human eye) · Resampling: intensity of a new pixel (bilinear) · Pseudo-code: decrease resolution 2:1 · Sketch the image resampled to lower ppi
☐	10:00–10:50	**Colour** Real exam task (EXAM 1/23): 3 objects → 3 bits, count (512), mark in the HSV cone. Grey: write BOTH weighted and average. Draw the RGB cube and the HSV cone once from memory. *Types:* How many colours for b bits per channel · Reduce colour/grey depth (8 → n bits) · RGB → CMY · RGB → greyscale · Mix two RGB colours · Mark colours in the HSV cone · Chromaticity diagram: what can be mixed · Describe the RGB model
☐	11:00–11:50	**Lines: DDA, Bresenham, antialiasing** Trace Bresenham by hand twice (table: current, p, decision, next). $p \geq 0$ goes up; start at the smaller x. Last 10 min: pipeline stages, fragment shader I/O, cross product. *Types:* DDA: steps and pixels · Why slope ≤ 45° in basic DDA · Bresenham: steps and pixels · Antialiased version of a line · Stages of the graphics pipeline · Fragment shader input and output · Vectors: components, sum, cross product, matrix × vector
☐	12:00–12:50	**Circle and fills** Circle r = 10 table from memory ($p_0 = 5/4 - r$, NEW x, y, y−1 when $p \geq 0$, mirror 8 ways). Scan-line: cross, sort, pair, fill. Seed fill: name the seed and the order L, R, U, D. *Types:* Midpoint circle · Scan-line fill · Seed (starting-point) fill: order of pixels
☐	12:50–13:40	**Lunch + walk, no screens** A real break. A short walk helps more than scrolling.
☐	13:40–14:30	**Barycentric + morning review** Inside test + grey value (official pixel (1,2) → 96; variant 10/25/128 → 47) + texture pixel (UV). Then 20 min of cards on "All", mode "Due + new": mixing the morning topics. *Types:* Barycentric: inside test and colour
☐	14:40–15:30	**2D transforms + frame change** Exam favourite. M = last·…·first. Do all four 2020 groups (translate then R45°) and scale about (–3,2). Frame rotated +θ = point rotated –θ → (2, –2). *Types:* Compose 2D transformations and apply · Change of coordinate frame
☐	15:40–16:30	**Clipping** Cohen–Sutherland on 3 segments (codes TBRL, AND ≠ 0 reject, AND = 0 ≠ accept), all four 2020 groups. Sutherland–Hodgman: write all 4 stages with the 4 cases. *Types:* Cohen–Sutherland line clipping · Sutherland–Hodgman polygon clipping
☐	16:40–17:30	**Bézier** Quadratic (1,1),(4,8),(10,1) and cubic (1,1),(3,–7),(8,–7),(10,1) at t = .25, .5, .75 by construction + drawing. Smooth join: C0, G1, C1 = $2P_3 - P_2$. *Types:* Bézier curve construction (+ smooth joining)
☐	17:30–18:15	**Dinner** Break. Drink water.
☐	18:15–19:45	**Full mock, real length** Mock tab → 90 min, printed pack allowed (like the real test). Then reveal and mark yourself honestly. Save the score.
☐	19:55–20:45	**Fix what you lost** Redo every task you lost points on, with new random numbers (Solve → Random numbers). Then 15 min of True/false cards.

✓	time	block
☐	20:45–21:15	**Annotate the pack + pack the bag** Write your own notes in the printed pack from memory (traps you made today). Bag: pack, squared paper, ruler, 2 pencils, eraser, pen, calculator if allowed, student ID.
☐	21:15	**Stop. Sleep before 23:00** Memory is consolidated in sleep. No new topics tonight.

Thursday 12 Nov · light (about 1 hour)

✓	time	block
☐	any time	**Due cards (20 min)** Cards tab, "Due + new" on All. Only what the app schedules.
☐	any time	**One 45-min mock** Mock tab → 45 min, new numbers. Fix only what you lose.
☐	evening	**Read the traps box, sleep 8 h** Pack page 2: "Traps" and "Wrong answers circulating". Bag ready at the door.

Friday 13 Nov · test day (08:30)

✓	time	block
☐	breakfast	**Warm-up 10 min** 10 cards + one Bresenham trace. No new material.
☐	in the test	**Order of work** Read all tasks first. Do the calculation tasks first (sure points). For every task: formula → steps table → drawing → boxed answer + your convention. Partial credit counts.

Mock scores: 08:30 pre-test ____ / 15 · 18:15 full mock ____ / 15 · Thursday ____ / 15 **Tasks I lost points on:**
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